pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Setup: Python, VS Code, and Colab

SETUP

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Two ways to take the course

Route	What you need	Parts to follow
Colab, in your web browser	A Google account. Nothing is installed.	Part 5 only
Your own Windows computer	Python and VS Code, installed here	Parts 1 to 4

Not sure? Start with Colab.

It is the quickest way in. Mac users: use Colab. You can set up your own computer later.

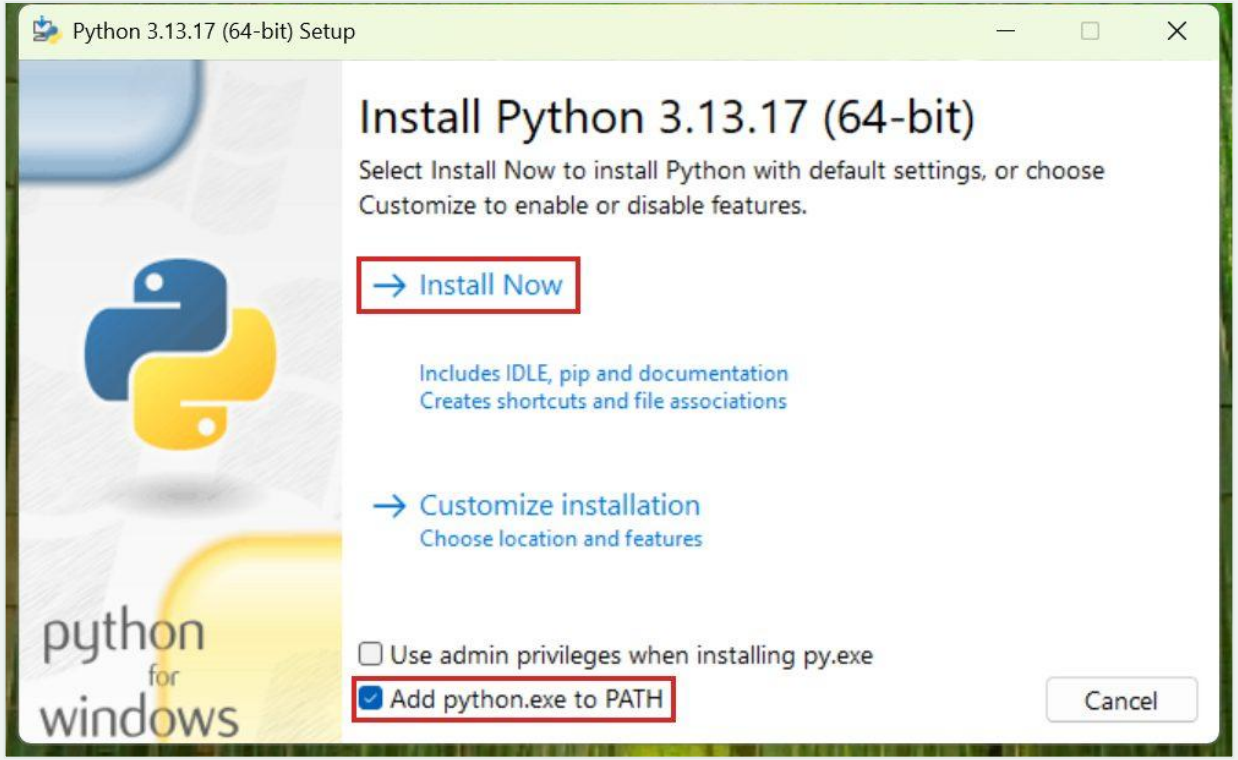
On a work computer

Check your firm's rules before installing anything. Never upload work files to Colab.

Text too small or too big? Ctrl and + makes everything bigger, Ctrl and - smaller. In VS Code and every browser.

Part 1: install Python

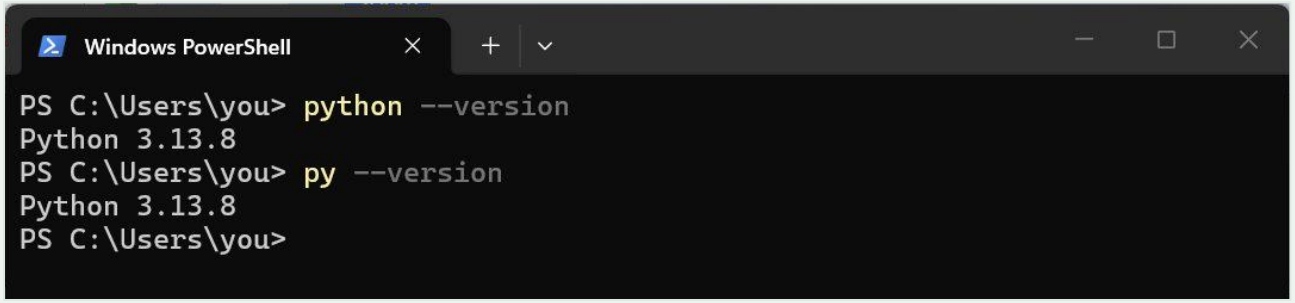
1. python.org/downloads/windows: under Stable Releases, the newest Python 3.13, Windows installer (64-bit).
2. Run it from Downloads.
3. Tick Add python.exe to PATH, at the bottom. Without it, Windows cannot find python or pip.
4. Click Install Now. Wait for Setup was successful, then Close.



Check that Python works

Start, type PowerShell, open Windows PowerShell, then type:

```
python --version
```



If you see	Do this
python is not recognized, or the Microsoft Store opens	Close PowerShell, open it again, retry
Still not found	Try py --version, and use py wherever the course says python
Neither works	Run the installer again with the PATH box ticked

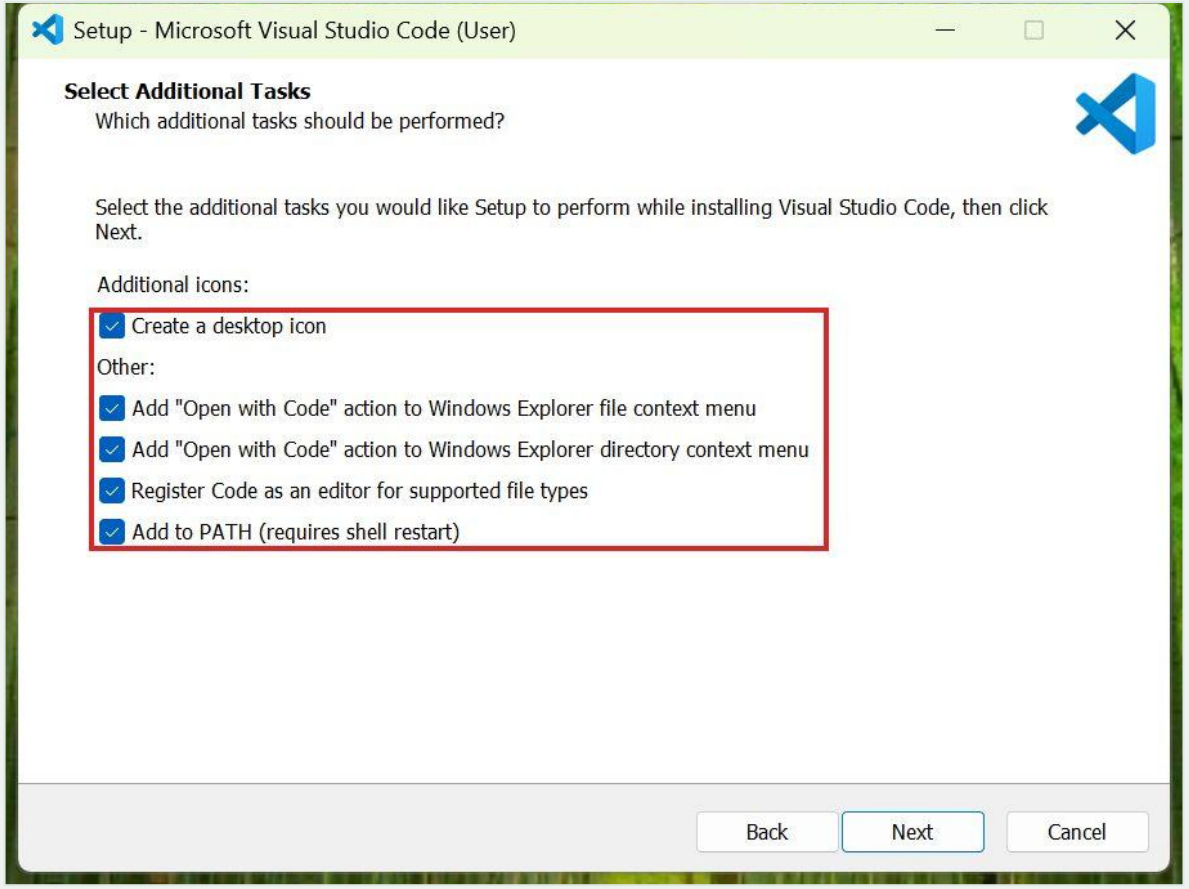
Part 2: install VS Code

1. code.visualstudio.com/download: click Windows.

2. Run it. Accept the license, Next. Keep the location pages as they are.

3. Extra tasks: tick all five, Create a desktop icon and the four under Other.

4. Copilot page, if shown: leave it, Next. Then Install, and Finish.

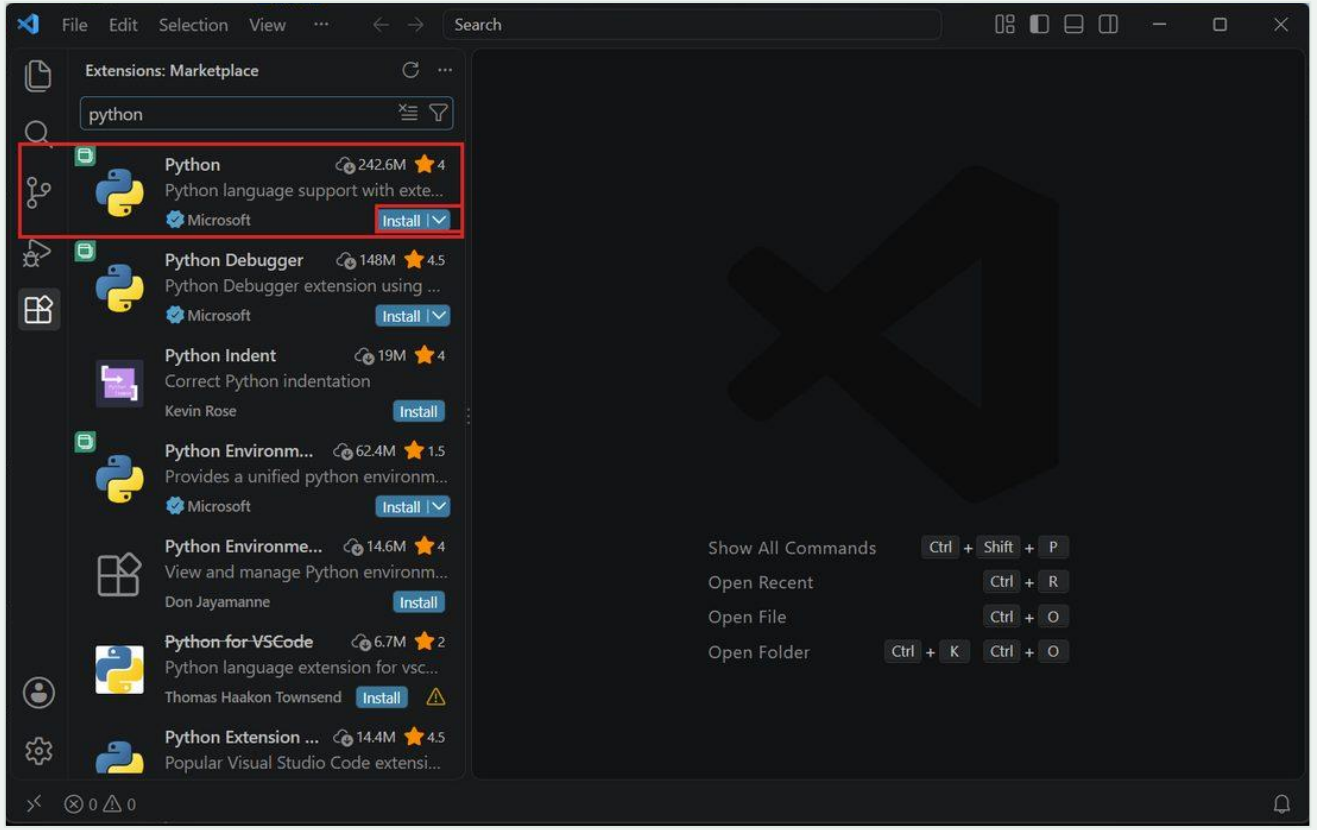


Two extensions: Python and Jupyter

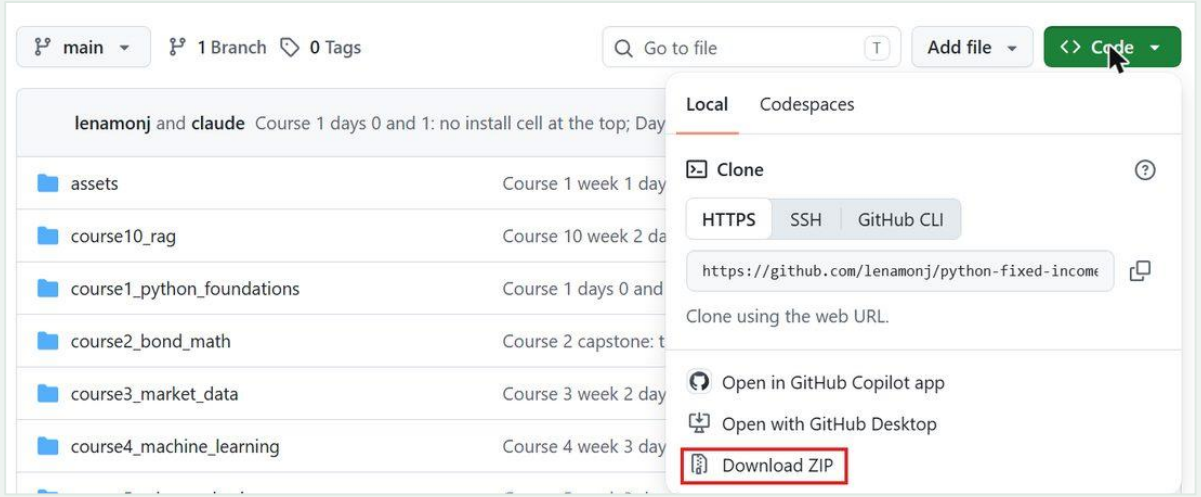
1. Click Extensions: the four small squares on the left (Ctrl + Shift + X).

2. Search python. Install Python, by Microsoft (blue tick).

3. Search jupyter. Install Jupyter, by Microsoft.



Part 3: get the course onto your computer



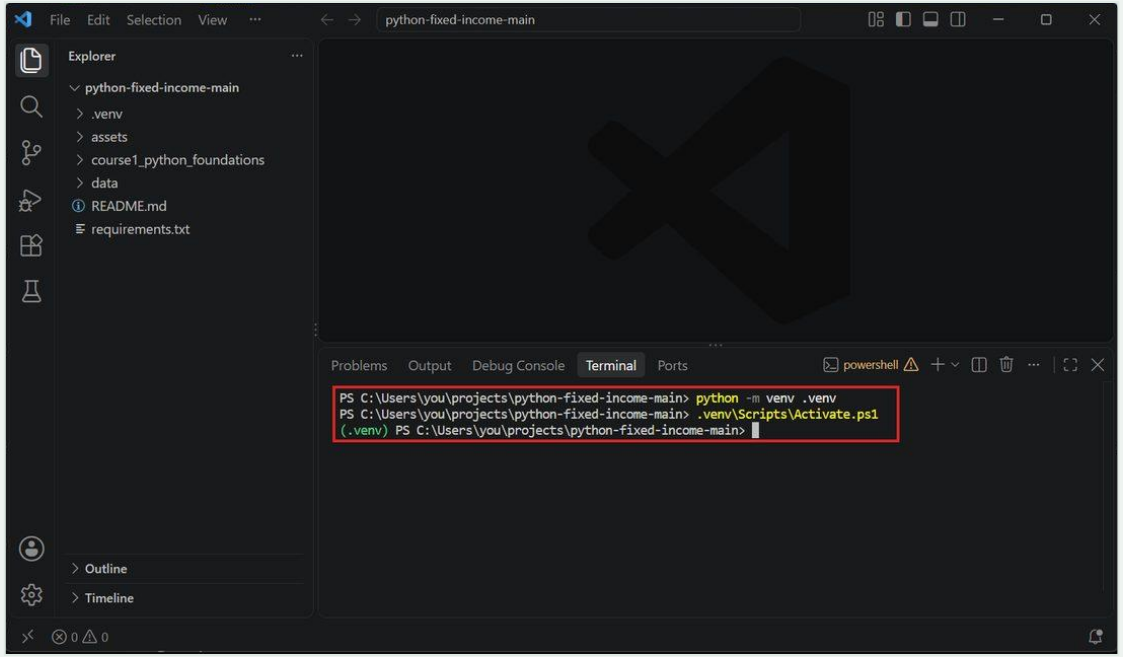
1. github.com/lenamonj/python-fixed-income: Code, then Download ZIP.
2. Make a folder called projects in your user folder.
3. Extract All the ZIP, and Browse to projects.
4. VS Code: File, Open Folder, python-fixed-income-main.

File Explorer: the yellow folder on the taskbar, or Windows + E. If VS Code shows only another python-fixed-income-main inside, open the inner one.

Part 4: the course environment

In VS Code: View, then Terminal. Type each line and press Enter.

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -r
course01_python_foundations\requirements.txt
```



Line 1 once per computer. The prompt then starts with (.venv). The install ends with Successfully installed.

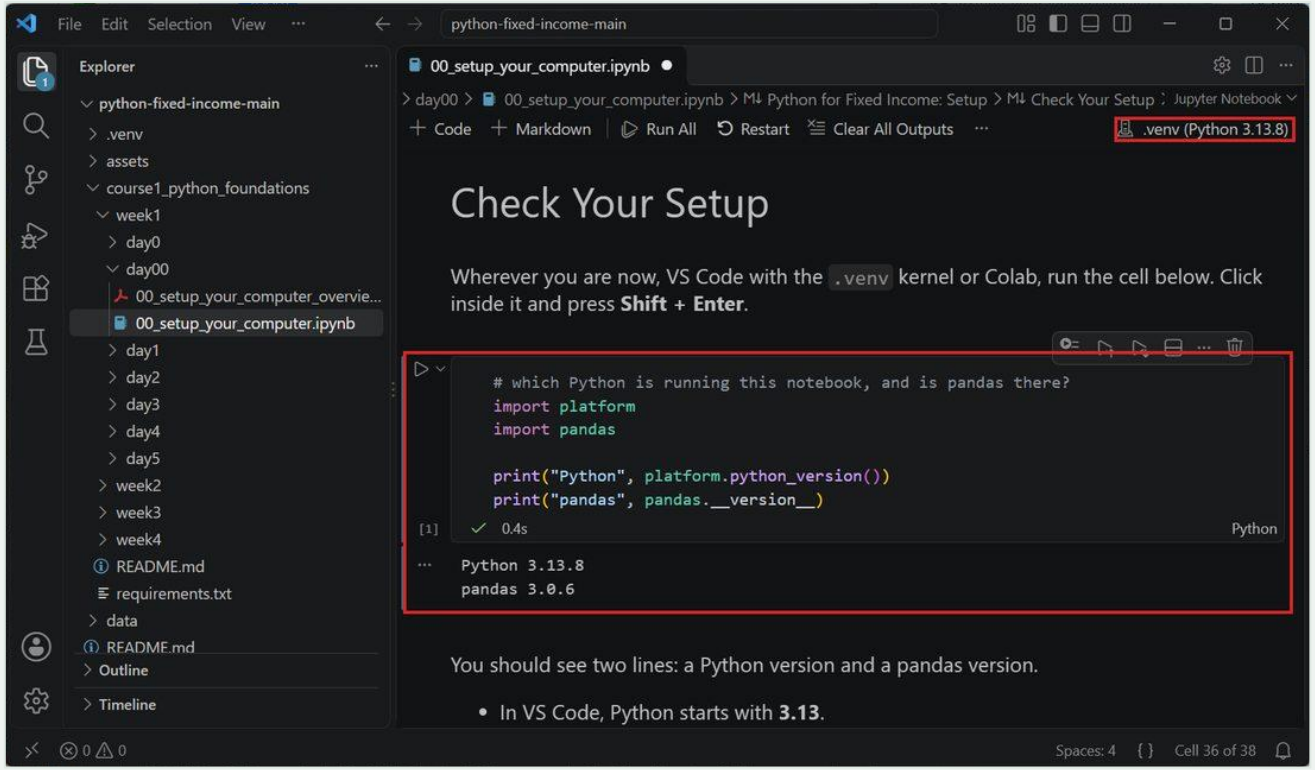
Choose the kernel and run a cell

1. Open setup, and this notebook, setup_your_computer.ipynb.

2. Top right: Select Kernel, Python Environments, the .venv line (Recommended).

3. Scroll to Check Your Setup at the end. Click in its code cell, Shift + Enter.

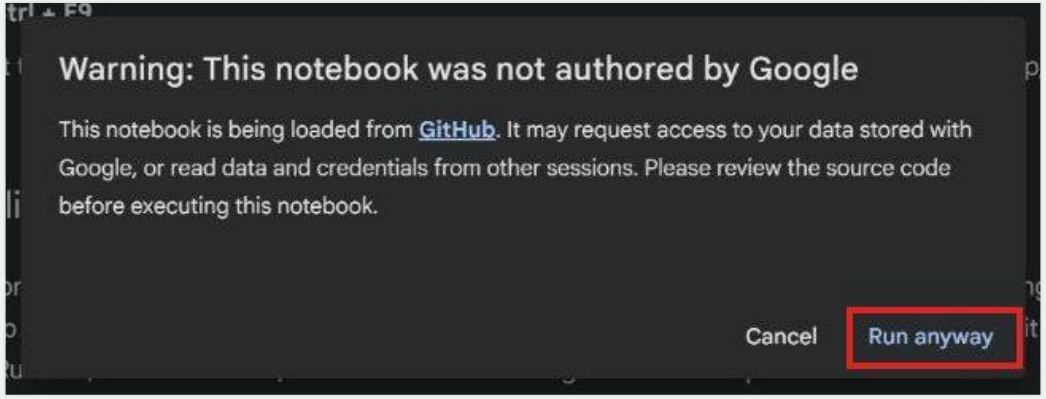
4. A Python 3.13 version and a pandas version appear. You are set up.



Shortcuts: Ctrl + and Ctrl - zoom. Ctrl + S saves. Shift + Enter runs a cell and moves on. Ctrl + Enter runs and stays. Ctrl + ~ opens the terminal. Ctrl + P opens any file by name.

Part 5: Google Colab

To	Do this
Open a course notebook	On GitHub, click its Open in Colab badge
Run a cell	Its round play button, or Shift + Enter
Run every cell	Runtime, then Run all (Ctrl + F9)
Keep your work	File, then Save a copy in Drive
Add a cell	Hover between two cells: + Code or + Text



First run: this warning appears for every notebook from GitHub. Click Run anyway.

Public or made-up data only in Colab, like the course's.

What you can do now

- Check Python with python --version
- Open the course folder in VS Code
- Switch on .venv and install the course's libraries
- Pick the .venv kernel and run a cell
- Open, run, and save a course notebook in Colab

Day 0 Start Here: what Python, a notebook, and a library are